

dunnhumby

Retail Store Performance Analysis

Through Customer Analytics and Targeted
Campaigns

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| TABLE OF CONTENTS

01

EXECUTIVE SUMMARY

02

PROJECT OVERVIEW

03

BUSINESS PROBLEM

04

DECISION TREE

05

METHODOLOGY

06

KEY INSIGHTS

07

FORECASTING

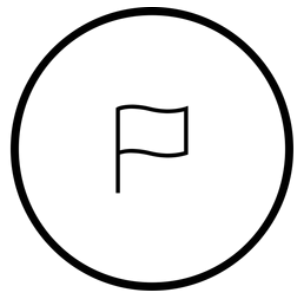
08

RECOMMENDATIONS

09

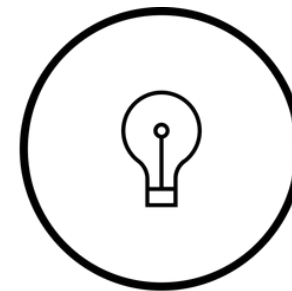
DASHBOARD

| EXECUTIVE SUMMARY



The Challenge

88% of stores are underperforming, with most growing below the monthly average. The top **12% of stores drive 80% of total sales**, creating a critical performance gap.



Solution

Deploy **Campaign 18** to a **high-impact** subset of underperforming stores with active Best Customers. Optimize delivery for **afternoon engagement (12pm–6pm)** to drive sales in underperforming stores.



Expected Impact

278.7%
Net Campaign ROI

+30.1%
Sales Uplift

\$41K
Incremental Revenue

*Results outperformed industry benchmarks by 1.2-2x (Sources: WARC, Dunnhumby, Umbrex).

| PROJECT OVERVIEW



What is **Dunnhumby**?

Business intelligence company and a global leader in customer data science and analytics, **experts in working with brands and retails.**



Dataset Brief

- Contains **2,500,000 household level transactions** who are frequent shoppers at a retailer.
- Covering data of a 2-year period of time, from **2017-2018**.

| BUSINESS PROBLEM

Store Performanace Gap

Total Stores: 582

Performing Stores: 71 (12%)

Underperforming Stores: 511 (88%)

Pareto Principle



12% of stores



80% of sales

Growth Rate Analysis

Average Monthly Growth 9.3%



Stores Below Average:

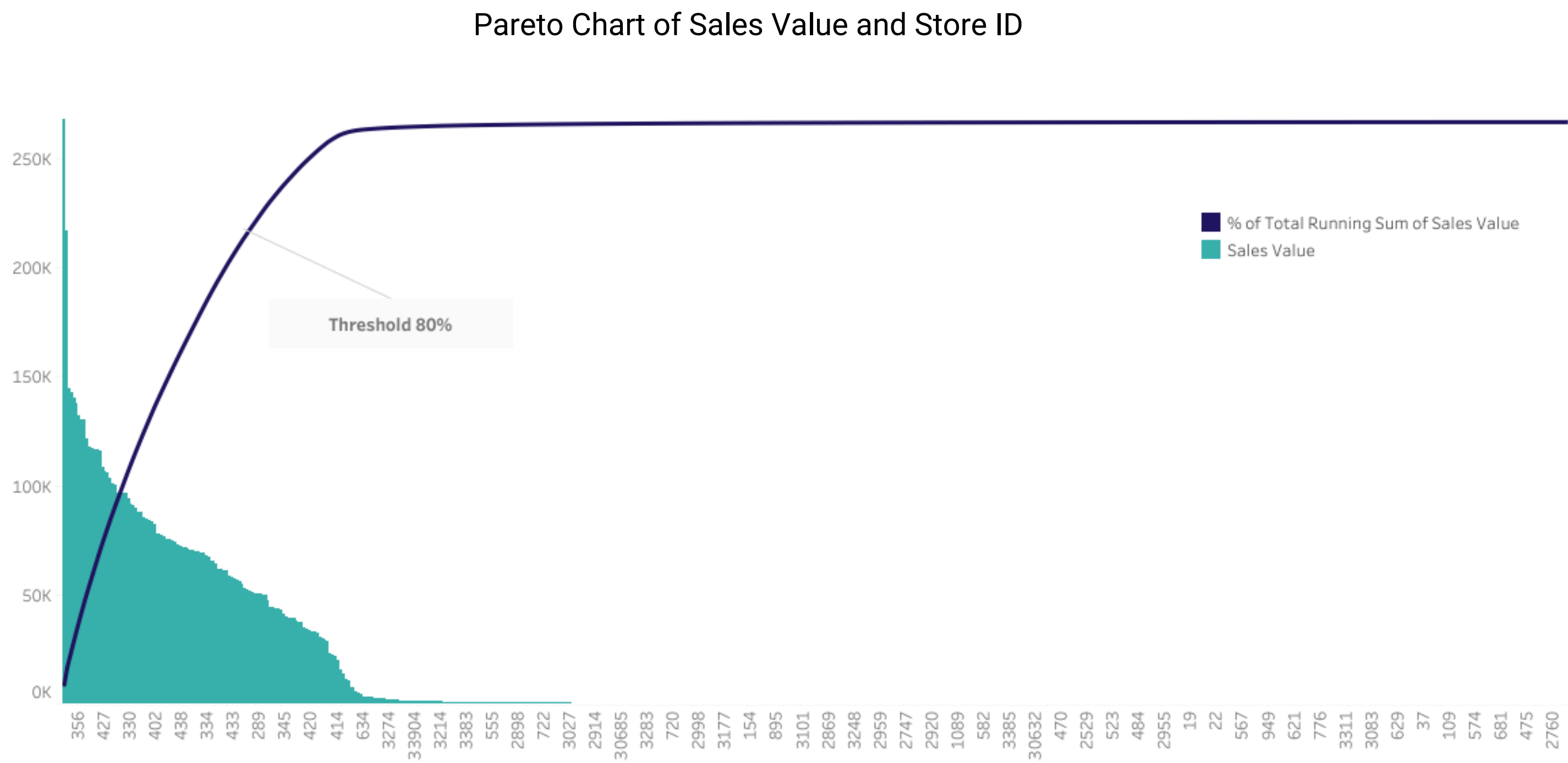
96.6%

of underperforming stores

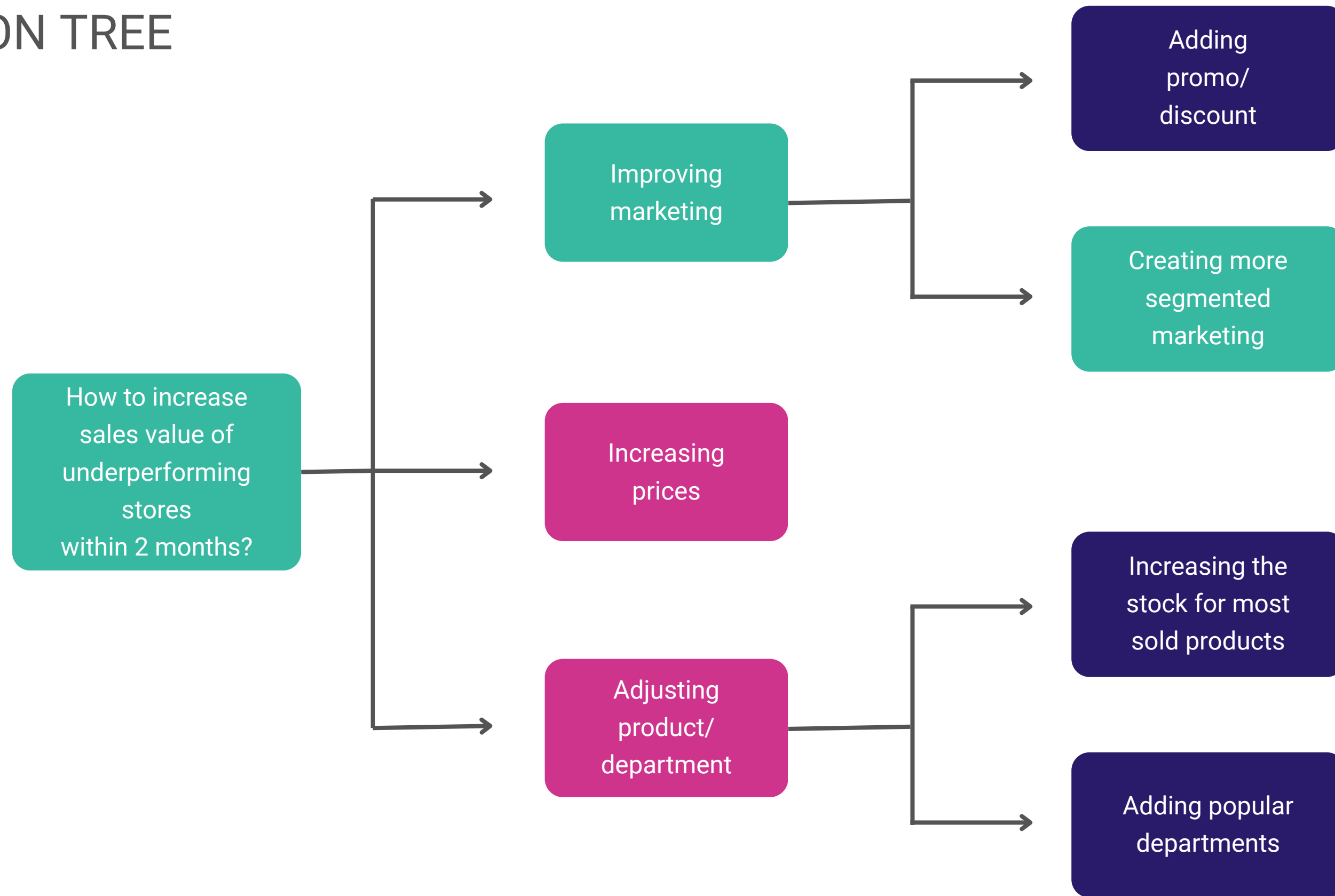
Business Question: How to increase sales value of underperforming stores within 2 months?

80% of Total Sales Driven by Just 12% of Stores

80% of sales value is driven by only 12% of 582 stores; the remaining 88% are underperforming, creating the critical sales gap.



| DECISION TREE



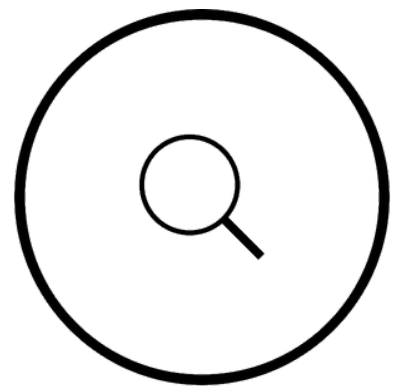
Underperforming stores must identify high-value customer segments and deploy targeted campaigns to accelerate sales value within the 2-month window.

| METHODOLOGY



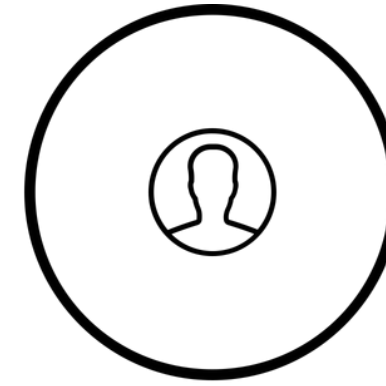
Data Acquisition & Feature Engineering

Collected over 2.5 million household-level transactions from Dunnhumby's retail database. Merged five datasets using key identifiers, handled missing values, and created temporal features for segmentation and forecasting.



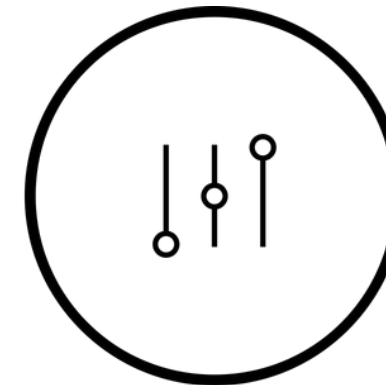
Campaign Efficacy Analysis

Compared historical campaign performance through ROI, conversion rate, and timing analysis. Validated engagement patterns using t-test to confirm statistical significance of sales uplift drivers.



RFM Segmentation & Audience Prioritization

Applied Pareto-based store segmentation and RFM modeling to classify stores and customers. Calculated Recency, Frequency, and Monetary scores to rank segments by campaign responsiveness.

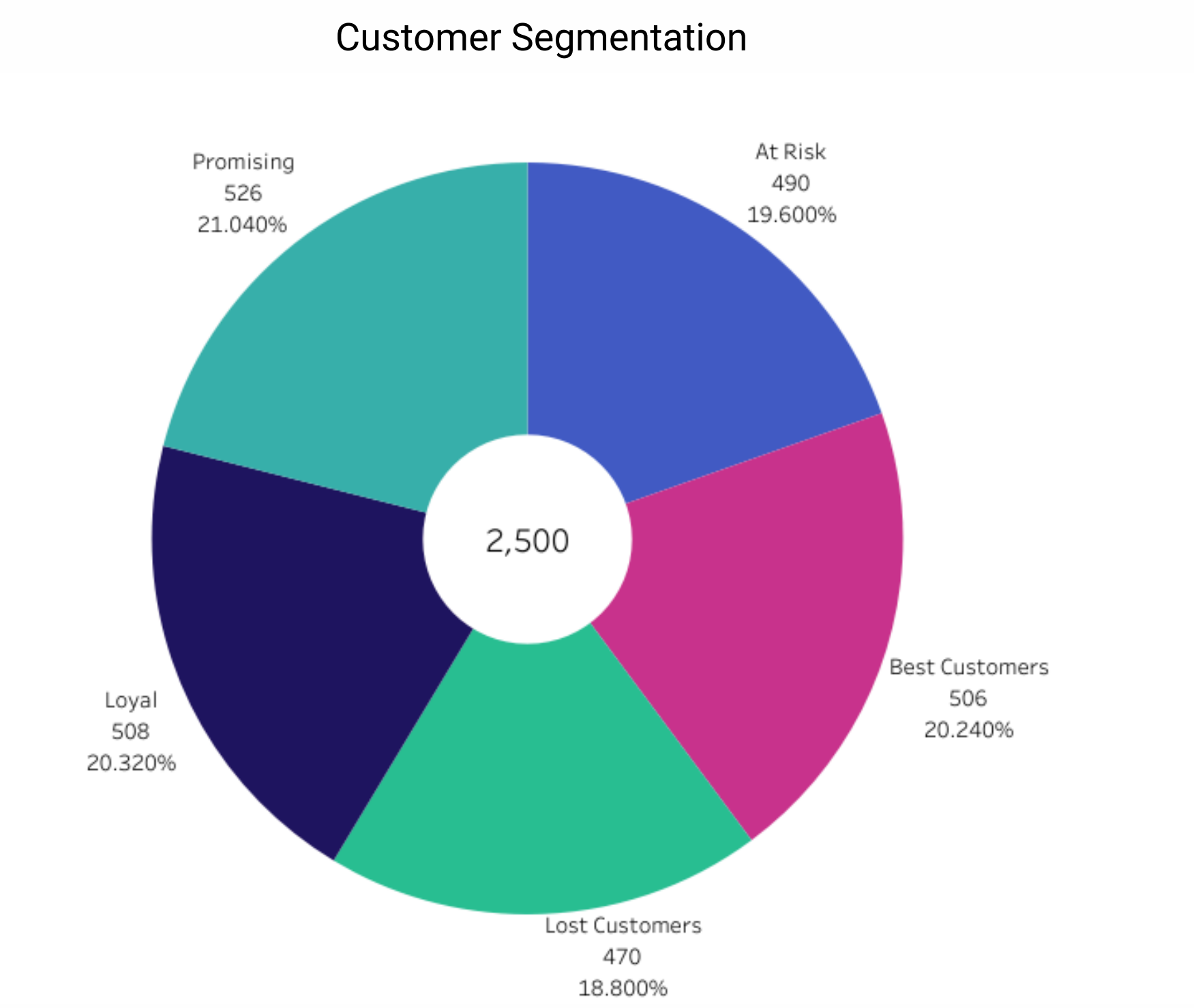


Predictive Sales Modeling

Developed a LightGBM gradient boosting regressor with train-test split validation. Utilized Optuna for Bayesian hyperparameter optimization to minimize prediction error and ensure model generalization.

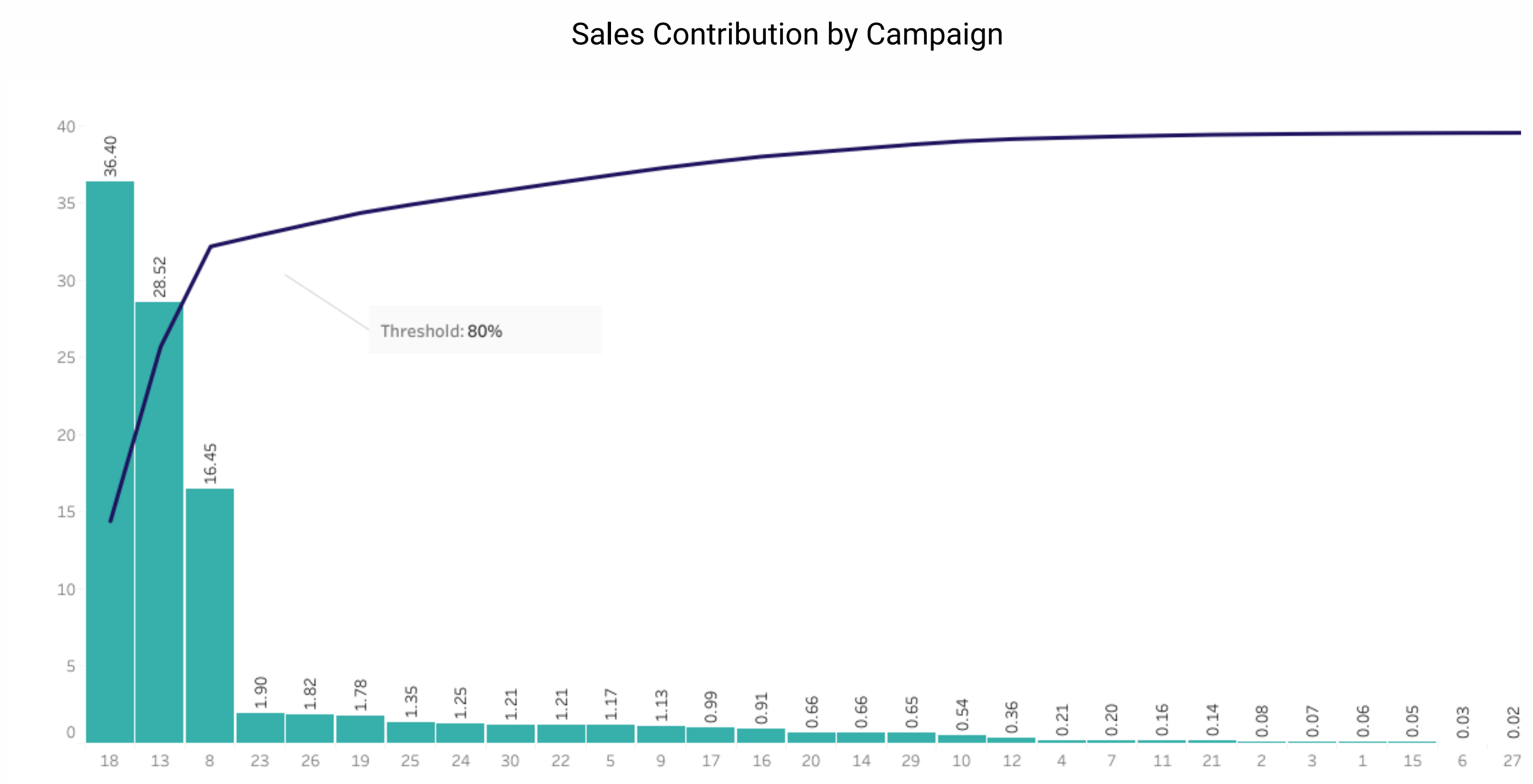
Promising, Loyal and Best Customer Segments Offer the Highest Growth Opportunity

RFM segmentation revealed that these groups show high purchase intent but lower recency, ideal for re-engagement.



Top 3 Campaigns Drive 80% of Total Sales

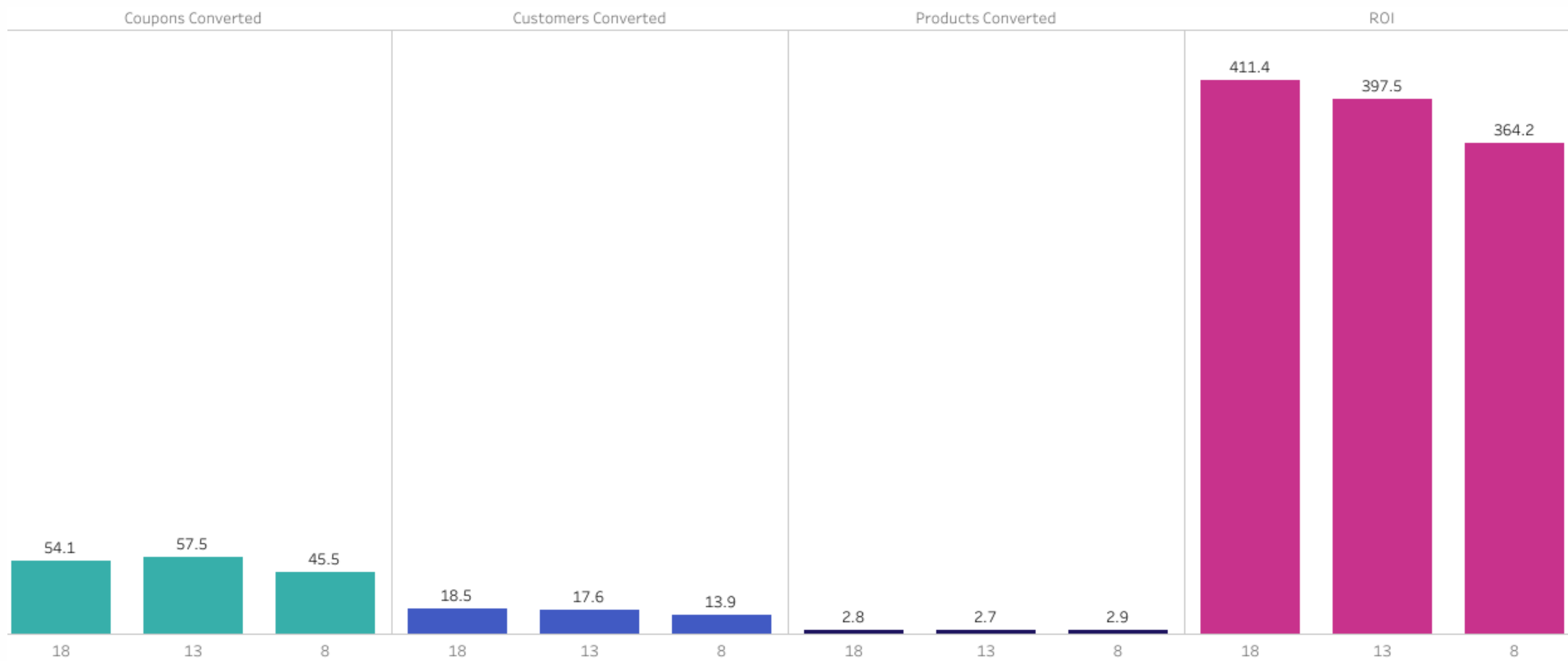
A small group of campaigns contribute most of the total revenue, highlighting where marketing focus should be placed.



CAMPAIGN 18 OUTPERFORMED ALL CAMPAIGNS WITH 278.7% NET ROI

Among top campaigns, Campaign 18 achieved the highest ROI and conversion rate, proving its scalability and impact.

Campaign Performance Comparison (ROI vs Conversion)



Note: Net ROI calculated after deducting 75% COGS and a \$1,500 campaign cost; Gross ROI before cost adjustments was 411.4%.

| CAMPAIGN 18 DESIGN & MECHANICS

1. OFFER STRUCTURE

- 15% discount on top 3 purchased categories
- \$25 minimum basket threshold
- 10-day validity period
- 2x loyalty points bonus

2. DELIVERY METHOD

- Digital vouchers (email + mobile app)
- Personalized messaging

3. WHY IT WORKS

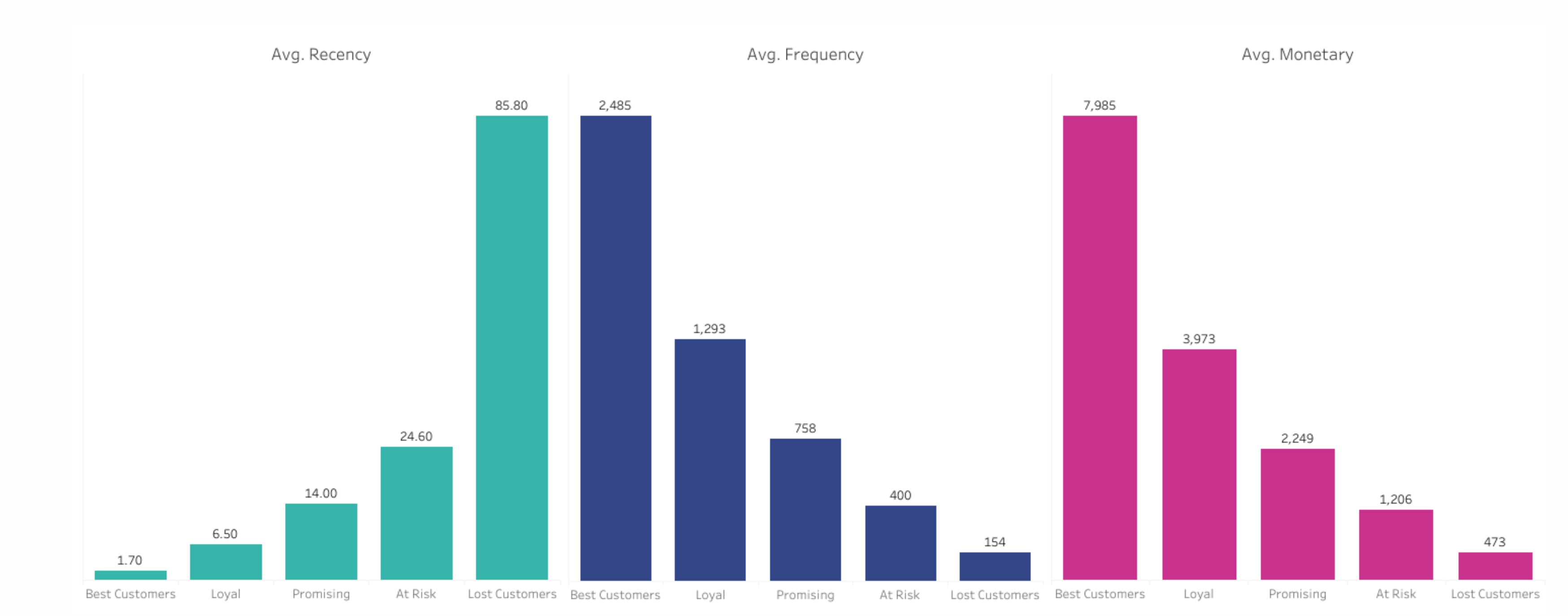
- Targets products customers already love
- Basket threshold protects margin
- Urgency drives immediate action
- Aligns with Dunnhumby's proven methodology



Best Customers Drive the Highest Sales and Engagement

Low recency, high frequency, and high spending make this segment the easiest to retain and most responsive to campaigns, ensuring the fastest sales impact.

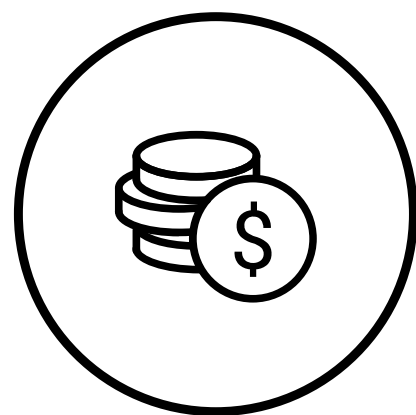
RFM-Based Customer Groups Overview



Middle-Aged, Dual-Adult Households with Moderate Income Lead Sales

Customers aged 45–54 with income between \$50K–\$74K and no kids form 24% of the active base, ideal for afternoon campaigns.

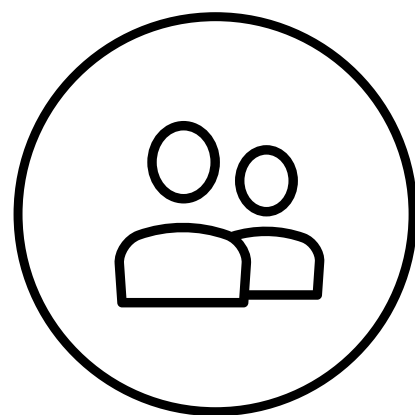
Demographic Breakdown of Best Customers



**Having income
50-74k**

24.65%

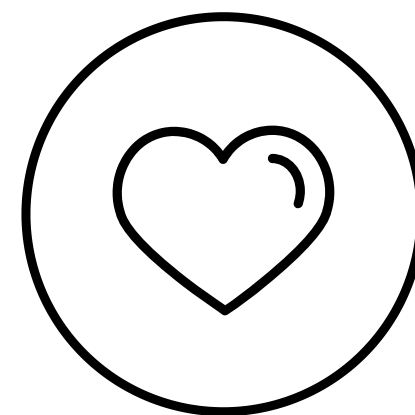
of total known demographics



**Two adults and
No kids**

29.75%

of total known demographics



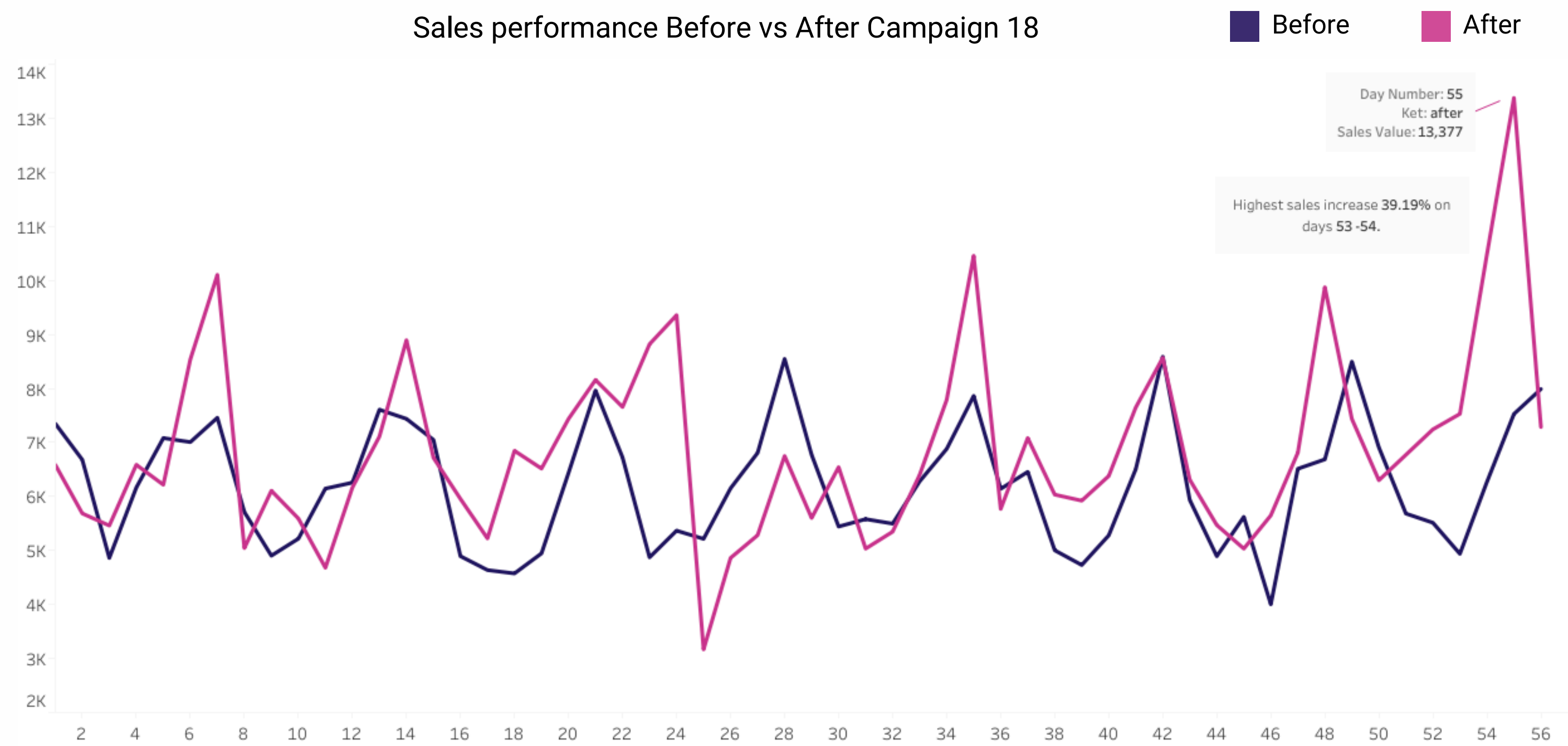
**45-54
years old**

34.45%

of total known demographics

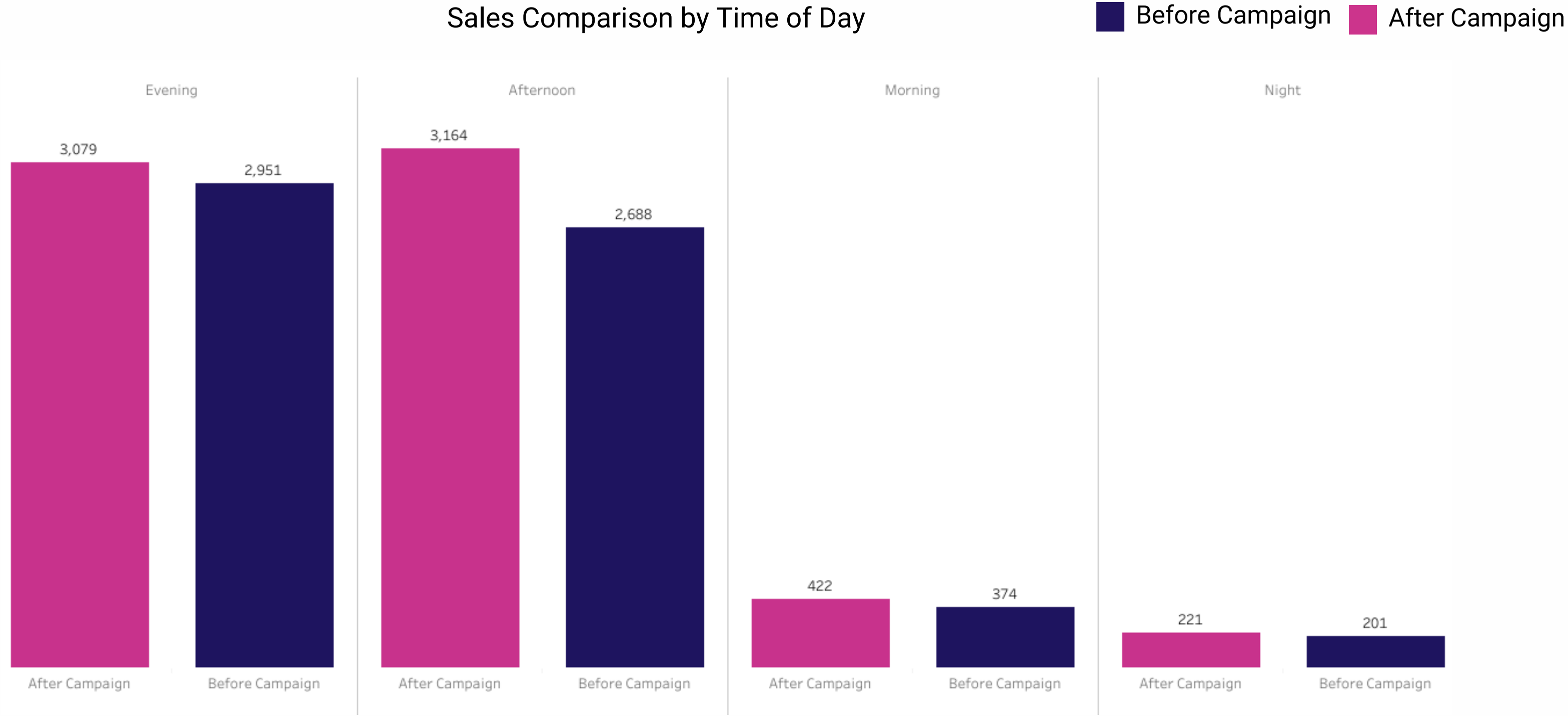
Campaign 18 Drove 10.8% Sales Growth in 55 Days

Targeting Best Customers through Campaign 18 increased sales by 10.8%, with +0.19% percentage points of daily growth acceleration.



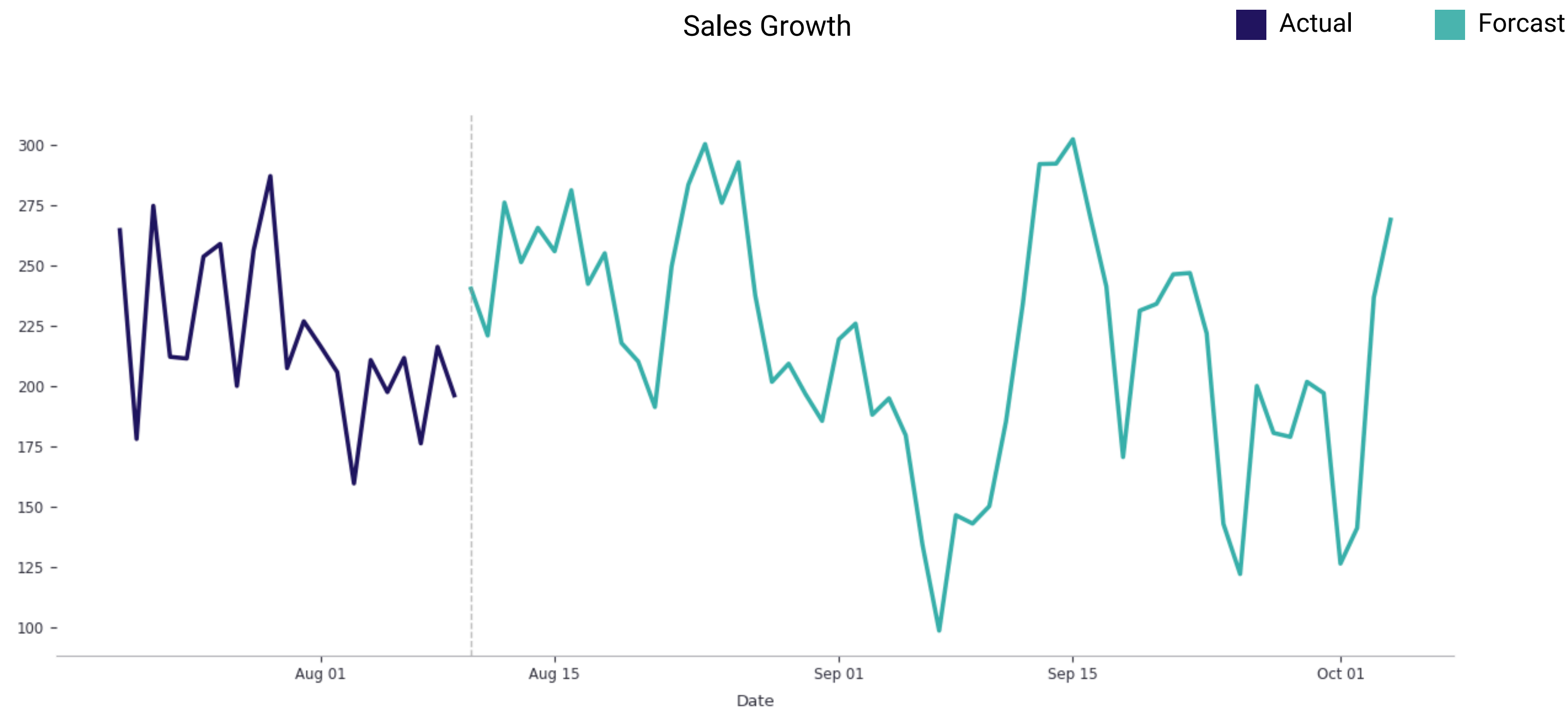
Afternoon Drives 17% Higher Average Sales and 27% of Total Sales

Customer engagement peaks in the afternoon, making it the most effective time to deploy campaigns.



VALIDATED 30.1% SALES UPLIFT FOR 3 TEST STORES IN 56 DAYS

Applying Campaign 18 during afternoon hours delivered a pooled sales uplift of 30.1% for store 299, 317 and 448. If scaled to all 85 eligible underperforming stores, this strategy could drive ~\$41K in incremental revenue.



Note: Validated through 2,000 statistical simulations; result held positive 99.9% of the time.

| WHY STORES 299, 317, 448?

The strongest pilot cohort among 511 underperforming stores, balancing recovery need with sufficient customer density and a stable sales history for reliable validation.

- **Underperforming by design:** Bottom 15% by sales, representing the recovery target group
- **Recoverable:** 37, 34, and 28 Best Customers respectively, the highest verified counts among qualifying underperformers
- **Validated:** Measurable uplift during the Campaign 18 window supported by 480 days of clean daily sales history



| RECOMMENDATIONS

- **Scale Campaign 18 to the Best Customer segment:** Target the highest-ROI campaign (278.7%) to the most responsive customer to drive measurable sales uplift.
- **Deploy exclusively during afternoon hours:** Focus delivery in the 12–6 PM window to align with peak Best Customer shopping behavior and maximize redemption.
- **Monitor Campaign Impact:** Enable real-time tracking through a live dashboard to identify underperforming stores and track customer response during rollout.

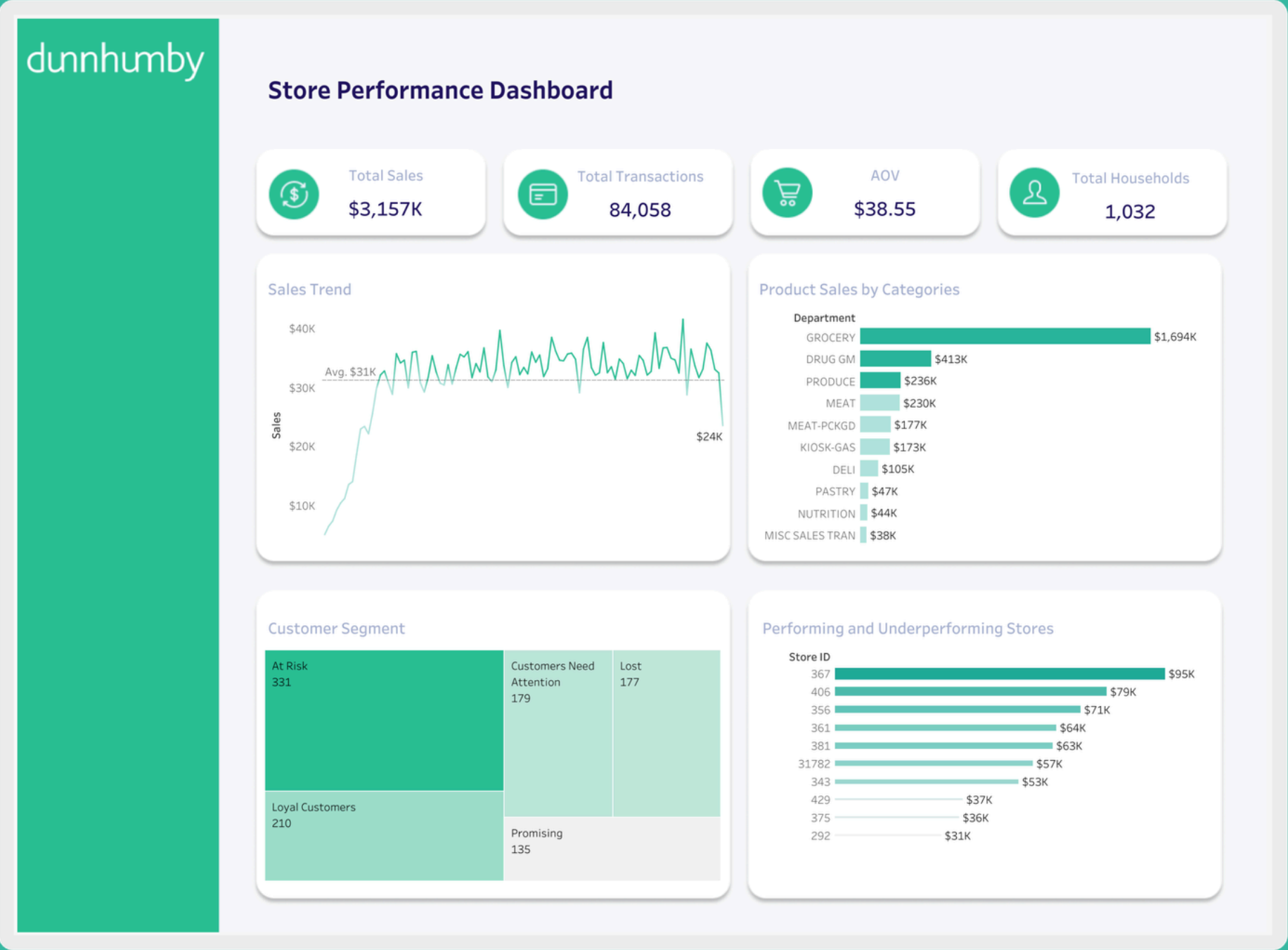


| ROLLOUT PHASE

Phase	Stores	Eligibility Criteria	Action	Timeline
Pilot Validation	05 (Tire - 1)	Top 5 Best Customer Count	Deploy Campaign 18 in afternoon	Week 1 - 4
Phase 1	25 (Tire - 2)	Next 25 Best Customer Count	Expand deployment	Week 5 - 8
Phase 2	55 (Tire - 3)	Next 55 Best Customer Count	Scale execution	Week 8 - 12
Excluded	426 (zombie stores)	Not Applicable	Do not target; no response possible	—

| DASHBOARD

Developed an interactive Tableau dashboard to monitor store KPIs, campaign performance, and sales trends, enabling stakeholders to identify underperforming stores and track campaign effectiveness in real-time.



Tableau

| PROJECT CONSTRAINTS & FUTURE SCOPE

- 1. Campaign Content Assumption:** The dataset provides Campaign IDs and performance metrics but no detail on the actual offer mechanics (discount %, products included, messaging). The recommended design strategy is therefore inferred from Dunnhumby established best practices.
- 2. Scaling Assumption Risk:** While the rollout is phased across 85 eligible stores, response may still vary by store profile. Continuous monitoring during Tier 1 validation will refine targeting for Tiers 2–3.
- 3. External Factors Excluded:** The LightGBM model's forecast does not account for external market factors (e.g., competitor activity, weather, local economic conditions) which could impact future sales performance.
- 4. Causality Validation is Required:** A retrospective difference-in-differences check found only a +1.2pp growth difference (95% CI crossing zero) between campaign recipients and untargeted customers, meaning true causal impact isn't yet confirmed. A formal A/B test (treatment vs. holdout) should be prioritized before further rollout investment.
- 5. Future Scope:** Operationalize the LightGBM model as a recurring forecasting engine to enable proactive, data-driven campaign targeting.

| REFERENCES

Industry Benchmarks

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Thank You

For any queries related to this project, feel free to connect me on the [Linkedin](#).